10.7 pH CALCULATION

The pH is the negative logarithm of the hydrogen ion concentration

$$\mathbf{pH = -\log_{10}[H^+]}$$

Example: Calculate the pH of 0.01 M HNO_3?

Solution:

$[H^+] = 0.01$

$pH = -\log_{10}[H^+]$

$pH = -\log_{10}[0.01]$

$pH = -\log_{10}[1 \times 10^{-2}]$

$pH = -(\log_{10}1 - 2\log_{10}10)$

$pH = 0 + 2 \times \log_{10}10$

$pH = 0 + 2 \times 1 = 2$

$pH = 2$

pOH: The pOH of an aqueous solution is realted to the pH.

The pOH is the negative logarithm of the hydroxyl ion concentration

$$\mathbf{pOH = -\log_{10}[OH^-]}$$

Example: The hydroxyl ion concentration of a solution is 1×10^{-9}M. What is the pOH of the solution?

Solution

$pOH = -\log_{10}[OH^-]$

$pOH = -\log_{10}[1 \times 10^{-9}]$

$pOH = -(\log_{10}1.0 + \log_{10}10^{-9})$

$pOH = -(0 - 9\log_{10}10)$

$pOH = -(0 - 9)$

$pOH = 9$

Relationship between pH and pOH

The pH and pOH of a water solution at 25°C are related by the following equation.

$$\mathbf{pH + pOH = 14}$$

If either the pH or the pOH of a solution is known, the other value can be calculated.

Example: A solution has a pOH of 11.76. What is the pH of this solution?

$$pH = 14 - pOH$$

$$pH = 14 - 11.76 = 2.24$$

10.8 PROBLEMS

Example 1: Calculate the pH of 0.001 molar solution of HCl.

Solution: HCl is a strong acid and is completely dissociated in its solutions according to the process:

$$HCl_{(aq)} \rightarrow H^+_{(aq)} + Cl^-_{(aq)}$$

From this process it is clear that one mole of HCl would give one mole of H^+ ions. Therefore, the concentration of H^+ ions would be equal to that of HCl, i.e., 0.001 molar or 1.0×10^{-3} mol litre^{-1}.

Thus, $[H^+] = 1 \times 10^{-3}$ mol litre^{-1}

$pH = -\log_{10}[H^+] = -\log_{10}10^{-3}$

$= -(-3 \times \log_{10}) = -(3 \times 1) = 3$

Thus, $pH = 3$

Example 2: What would be the pH of an aqueous solution of sulphuric acid which is 5×10^{-5} mol litre^{-1} in concentration.

Solution: Sulphuric acid dissociates in water as:

$$H_2SO_{4(aq)} \rightarrow 2\,H^+_{(aq)} + SO_4^{2-}{}_{(aq)}$$

Each mole of sulphuric acid gives two mole of H^+ ions in the solution. One litre of H_2SO_4 solution contains 5×10^{-5} moles of H_2SO_4 which would give $2 \times 5 \times 10^{-5} = 10 \times 10^{-5}$ or 1.0×10^{-4} moles of H^+ ion in one litre of the solution.

Therefore,

$[H^+] = 1.0 \times 10^{-4}$ mol litre^{-1}

$pH = -\log_{10}[H^+] = -\log_{10}10^{-4} = -(-4 \times \log_{10}10)$

$= -(-4 \times 1) = 4$

Example 3: Calculate the pH of 1×10^{-4} molar solution of NaOH.

Solution: NaOH is a strong base and dissociates in its solution as:

$$NaOH_{(aq)} \rightarrow Na^+_{(aq)} + OH^-_{(aq)}$$

One mole of NaOH would give one mole of OH^- ions. Therefore,

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